

Learn Programming by Making Video Games

Course Syllabus - 2026/27 School Year

VIDEO GAME CREATION

SLHE Homeschool Co-op • Instructor: Robert Kuropkat • robert.kuropkat@gmail.com

Course Description

This course teaches programming fundamentals through hands-on game development. Students choose a game engine or platform that fits their device and interests, then build increasingly capable games across a full school year, culminating in a finished, shareable project for the co-op's CoOp Expo. Rather than locking the class into a single tool, each week's core lecture rotates through a different supported engine - Scratch, Godot, Unity, GameMaker, GDevelop, TIC-80, 001 Game Creator, and (optionally, for capable hardware) Unreal Engine - while providing working example code in every supported engine for every lesson. Students may switch platforms at any point. Note: 001 Game Creator's editor is Windows-only, so Mac/Chromebook students will use a different engine during weeks it's featured (see the Engine & Platform Guide).

Meeting Schedule

- **Format:** 2 hours per meeting: approximately 1 hour lecture/demo, 1 hour hands-on lab
- **Frequency:** Weekly, 30 total meetings across the school year
- **Semester 1:** 15 meetings, beginning Thursday, August 13, 2026
- **Semester 2:** 15 meetings, concluding with CoOp Expo project preparation
- **Homework:** Light take-home practice is expected between sessions, especially finishing in-lab exercises and installing/updating software

Audience & Prerequisites

Open to Middle and High School students. No prior programming experience is required. Because the class spans a wide age and maturity range, lessons are designed with flexible depth: younger or newer students can lean on block/visual coding (Scratch, GameMaker Visual, GDevelop, 001 Game Creator) and simpler project scope, while older or more experienced students are encouraged toward text-scripted engines (Godot, Unity, TIC-80, or Unreal) and more ambitious builds. Everyone works toward the same learning objectives each week at a level appropriate to them - see Assessment Approach below.

What You'll Need

A personal laptop (Bring Your Own Device) meeting the minimum specifications for your chosen engine, and a way to save/back up project files (USB drive or cloud storage). Full hardware guidance, install instructions, and a device-fit chart are in the companion Engine & Platform Guide, which is sent home before Lesson 1 so installation can begin ahead of time.

Curriculum Alignment

Course content is organized around the IGDA Game Education SIG's Curriculum Framework for the Study of Games and Game Development, adapted for a Middle/High School BYOD setting. The framework's most recent published version is v3.2 beta (2008); the IGDA Game Education SIG is actively revising it as of 2026, with a second public comment period running May 30-September 1, 2026 and publication targeted for September 30, 2026. This syllabus will be reviewed against the updated framework once published, likely partway through Semester 1.

Where a student's chosen engine is Unity, lesson content also maps to Unity's free Unity Learn Pathways - Unity Essentials, Junior Programmer, and Creative Core - so that a motivated student can work toward an official Unity Certification alongside class material.

Semester-by-Semester Overview

A week-by-week breakdown, including which engine is featured each week, lives in the companion 30-Lesson Master Schedule. The arc below shows the shape of the year:

Weeks	Focus	Milestone
1	Orientation: internet safety, game dev overview, platform selection	Engine installed & first project opened
2-6	Programming foundations: variables, events, loops, conditionals, input	First moving, controllable character
7-11	Core mechanics: collision, scoring, win/lose states, simple UI	Small playable prototype
12-15	Game feel & polish: audio, basic animation, level structure	Semester 1 capstone: complete Mini-Game
16-20	Intermediate systems: state machines, enemy/AI behavior, physics	Prototype with opponents/obstacles
21-25	Expanding scope: menus, save data, multiple levels, juice/polish	Feature-complete draft of capstone game
26-28	Playtesting, bug-fixing, presentation prep	Playtested, stable build
29-30	Finalize, package, and export shareable copies	Shareable build ready for CoOp Expo

CoOp Expo Capstone

The final weeks of Semester 2 (roughly Lessons 26-30) shift from new material to finishing, playtesting, and packaging each student's game so it can be demonstrated at the co-op's Expo and shared with friends and classmates. Students will export a runnable build (or shareable link, for browser-based engines), prepare a short demo/pitch, and get structured playtesting feedback from classmates before the Expo. The exact Expo date is not yet set for 2026/27; once it is confirmed, these final lessons may shift earlier or later in the 30-Lesson Master Schedule to line up with it.

Assessment Approach

This is a project- and mastery-based course, not a test-driven one, and it does not use a traditional numeric grading rubric. Because students span a wide range of ages and prior experience, every lesson's hands-on work - and both semester milestones - are structured around three levels of accomplishment instead:

Level	Think of it as...	What it means
Level 1 - Minimal	MVP (Minimum Viable Product)	The smallest working version of that lesson's concept - enough for the student to understand it and move on to next week's lesson. This is a full, legitimate completion of the lesson, not a fallback.
Level 2 - Complete	Internal Alpha/Beta	All of the lesson's expected features are present and working end-to-end. Art, polish, and final balancing may still be missing, and minor bugs may remain.
Level 3 - Refined	Ready to Publish	Feature-complete, tested, and polished enough that the student would be comfortable sharing it outside of class.

Reaching Level 1 is a complete, legitimate finish for that lesson and is always enough to move on to the next one - no student is ever stuck waiting to "catch up." Level 2 and Level 3 are open invitations for students with more time, skill, or interest to go further, not extra assignments or busy work. Each lesson's Student Handout spells out what Level 1, 2, and 3 look like for that specific exercise, and the Semester 1 Mini-Game and Semester 2 Expo game are evaluated the same way.

Policies

Internet & Digital Safety

Lesson 1 opens with an internet-safety unit grounded in NetSmartz materials, covering safe sharing, online interaction with strangers (including in multiplayer/online game contexts), and responsible use of school/lab wifi. Digital citizenship expectations are revisited whenever a lesson touches online accounts, multiplayer features, or publishing/sharing a game.

BYOD Expectations

Students bring a personal laptop meeting the minimums in the Engine & Platform Guide. Please charge devices before class and bring a charger - lab time is hands-on. Install large engines at home ahead of time when possible; shared classroom wifi is not fast enough for 20 students downloading Unity or Unreal simultaneously.

Engine Choice

Students select their own engine/platform from the supported list (see Engine & Platform Guide) and may change engines later in the year without penalty - every lesson provides working examples across all supported engines. Note: 001 Game Creator requires Windows, so Mac/Chromebook students should choose a different engine.

AI-Assisted Coding

Because this course sits at the intersection of programming education and modern development practice, thoughtful use of AI coding assistants is permitted and, in upper-level lessons, taught directly as a skill - but students must be able to explain what their code does and may not submit AI output they don't understand. Specific per-assignment guidance will be given as needed.

Academic Integrity

Collaboration and helping classmates debug is encouraged during lab time. All capstone work should represent each student's own code and implementation, with credit given for any tutorials, assets, or AI assistance used. For the Semester 1 Mini-Game, the game design itself is the same for the whole class (see "This Semester's Game" in your Lesson 1 Game Design Document) - what's yours is your code, your art/sound choices, and how far into Level 3 you push. The Semester 2 Expo game is fully each student's own design as well as their own code.

Attendance & Make-Up Work

As a homeschool co-op class, families manage their own attendance policies, but because lessons build week-to-week, missed lab time should be made up at home before the next session. Lesson materials and recordings of key demonstrations (where practical) will be made available.

Conduct

Standard co-op classroom conduct applies during lecture and lab time. Respectful, patient help for classmates using a different engine than you is part of the culture of this class.

Materials & Companion Documents

- **Engine & Platform Guide:** engine comparison, hardware fit, and install instructions - send home before Lesson 1
- **30-Lesson Master Schedule:** week-by-week topics, featured engine rotation, and IGDA framework tie-ins
- **Lesson Production Standards:** defines the three-level accomplishment framework and the standard set of materials produced for every lesson
- **Engine Installation & Hello World Verification Guide:** detailed setup instructions and a working test project for each supported engine, for instructor prep
- **Per-Lesson Materials:** for each lesson: a 1-page lesson plan, a teacher's script, a student handout per engine, and a teacher's prep checklist
- **Student Resource List:** supplemental tutorials, references, and practice sites by engine

Contact

Questions about the course, engine choice, or make-up work can be directed to Robert Kuropkat at robert.kuropkat@gmail.com.